Cloud Integrated Biometric Thumb Attendance System

Shreyansh Chougule, Kishor Edage, Alok Hase, Rushikesh Fartade

Trinity College Of Engineering & Research, Pune.

Department of Electronics and Telecommunication

Abstract

Biometric systems have found wide application in the contemporary field of operation to offer secure and precise identification. In this research work, the design and implementation of a cloud-enabled biometric attendance system are presented based on an ESP32 platform. The system makes use of a fingerprint module to recognize the users' identity and transfer the attendance data to a Google Sheet in real time wirelessly.

The system guarantees precision in attendance recording, eliminates fraud entry, and offers a cost-effective solution. Other attributes include an OLED screen and buzzer among others.**Keywords**

Biometric Attendance, ESP32, Fingerprint Sensor, IoT, Google Sheets, Cloud Storage, OLED Display, Wireless Communication

I. Introduction

Attending management is among the vital tasks performed in learning institutions and organizations. Manual methods have been proven to be inefficient in handling attendance and are likely to lead to mistakes like attendance via another person's name. Biometric solutions can be used as an effective solution for overcoming the challenges associated with manual solutions. Fingerprint recognition technology offers the highest level of security and uniqueness for all individuals.

This project will entail developing a cloud-based biometric attendance management system based on the ESP32 development platform. Data collected will include verifying the fingerprints and uploading attendance data to Google sheets.

II. **Literature survey**

1. Ezeofor Chukwunazo Joseph and Georgewill Onengiye Moses (2019) developed an IoT-based attendance monitoring system using fingerprint sensors and ESP32. The system provides accurate attendance and cloud storage but is limited by network dependency and system complexity.

2. Kotramma T. S. et al. (2023) proposed an automatic attendance system using machine learning and facial recognition techniques. The system improves automation and accuracy but requires high computational power and proper lighting conditions.

3. Jawaaz Ahmad and Zahid Gulzar Khaki (2016) presented a smart classroom attendance system using sensor-based technologies such as NFC and RFID. The system improves efficiency but lacks strong authentication, allowing possible proxy attendance.

4. Sudip Misra, Anandarup Mukherjee, and Arijit Roy (2021) explained the fundamentals of IoT, highlighting its role in enabling real-time data communication and cloud integration. However, IoT systems depend on reliable network connectivity and security mechanisms.

5. Research on biometric attendance systems indicates that fingerprint recognition provides higher accuracy and security compared to traditional systems. However, sensor performance may be affected by environmental conditions such as dirt or moisture.

6. Studies on RFID-based attendance systems show that while they are easy to implement and cost-effective, they are vulnerable to misuse since RFID cards can be shared among users.

7. Cloud-based attendance systems allow real-time data access and centralized storage, improving data management. However, they introduce concerns related to data privacy and internet dependency.

8. Embedded system-based attendance solutions using Arduino and ESP modules are widely used due to their low cost and simplicity. However, they may lack scalability and advanced processing capabilities.

9. Machine learning-based attendance systems improve automation and reduce human intervention, but they require large datasets, higher processing power, and complex implementation.

10. From the above studies, it is observed that no single system provides complete security, real-time access, and efficiency. Therefore, there is a need to develop a system that integrates biometric authentication with IoT and cloud technology to overcome these limitations.

III. System Architecture

1. Block Diagram

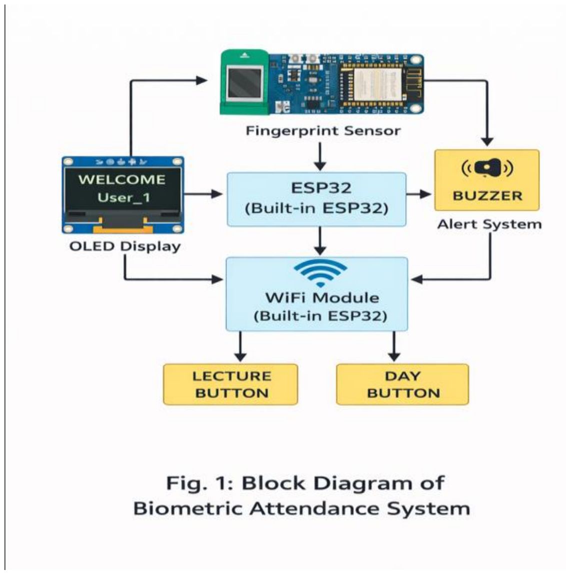


Fig. 1. Block Diagram of ESP-Based Biometric attendance System

This is a block diagram of the system, which demonstrates the functional arrangement of the cloud-based biometric attendance system.

ESP32 is the processing unit of this system. All actions of the system depend on this microcontroller.

Fingerprint reader is utilized in the biometric system for authentication purposes; hence when one places their finger on the sensor, their fingerprint gets scanned and transferred to the main microcontroller, where it is matched with the fingerprints stored in the system.

OLED screen shows relevant information about attendance like "Place Finger", "Welcome" and current attendance state.

Buzzer works as an alarm; it gives a sound whenever there's any mistake done while marking attendance, that is, when there is no valid fingerprint, and a person attempts to mark attendance.

ESP32 includes a Wi-Fi module; thus, it makes it possible for the system to connect to the internet and upload the attendance data collected into Google sheets; therefore, the attendance data, name, id, date, and time can be uploaded automatically.

There are two push buttons provided in the attendance system; one push button is to mark attendance for a new lecture and another push button is to mark attendance for a new day.**Flowchart of System Operation**

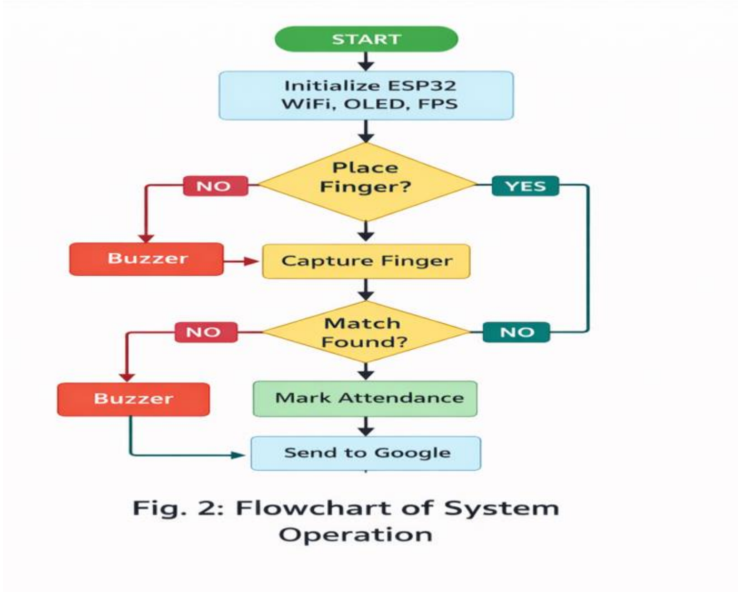


Fig. 2. Flowchart of System Operation

This flowchart illustrates the entire functioning mechanism of the biometric attendance system.

The whole process starts from the system initialization stage wherein all required hardware components like ESP32, WiFi, OLED display, fingerprint sensor will be initialized by ESP32 Microcontroller.

After initialization, the system goes to check whether there is any input from the user or not. For this purpose, the system waits till the user provides his/her input via placing their finger on the fingerprint sensor.

In case the input is not provided, the system again goes to the waiting stage, but if the system faces any error regarding the input or timeout then in that situation, the buzzer will work.

Once the input is taken, the system will detect the fingerprint and compare it with the existing templates in the database.

If the fingerprint does not match, then the buzzer will start working and the system will return to its initial state for new inputs.

In case if the fingerprint matches with the template, the system marks the attendance and the identification of the person will be done.

In the last step, the attendance information will be transferred to Google Sheets using WiFi module in ESP32.

IV. Circuit Design and Implementation

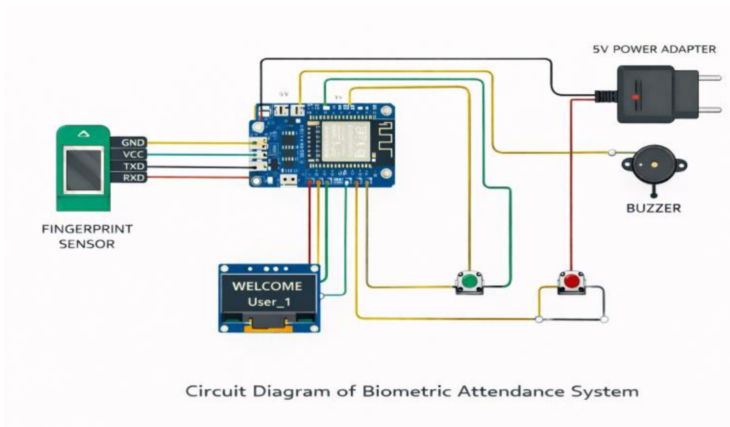


Fig. 3. Circuit Diagram of ESP32-Based Biometric Attendance System

The circuit diagram is the full implementation of the hardware design of the biometric attendance system with the help of ESP32 microcontroller.

ESP32 is the main microcontroller that performs all the functions of this attendance system by controlling all connected devices. It processes the fingerprint images, displays the output messages on the screen, takes input from the user, and also uploads data to the cloud server using the integrated WiFi module.

Fingerprint sensor is connected to the ESP32 microcontroller through the UART interface. Fingerprint sensor has four connections: VCC & GND for providing power supply, and TXD & RXD pins for communication.

OLED display is connected to the ESP32 microcontroller through the I2C protocol. OLED display uses four connections: VCC, GND, SDA, and SCL.

A buzzer has been interfaced with the ESP32 board to give out audio signals. This buzzer is triggered when a wrong finger is entered by the user or when he or she tries to take more than one attendance marks at once.

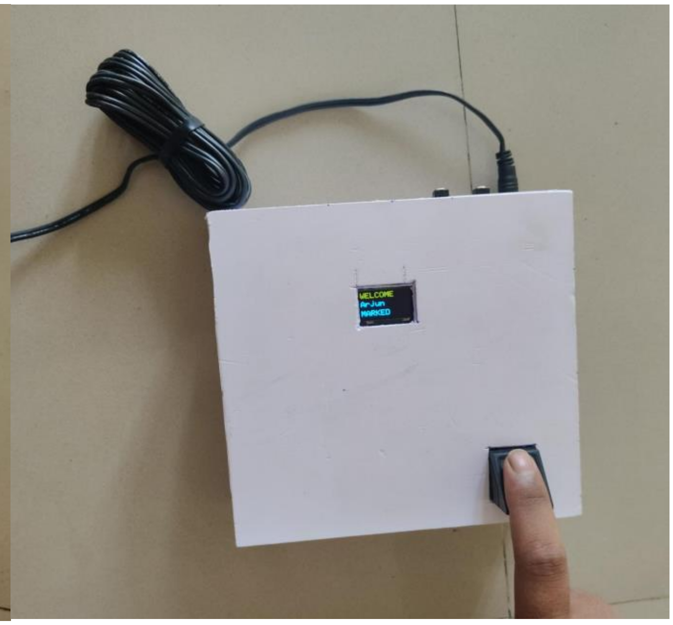
Two push buttons are interfaced with the ESP32 board. They are responsible for resetting the attendance of different sessions like lecture session or daily session. The connection of both push buttons to the ESP32 board is made from GPIO pins to ground pin with internal pull-up resistor on the ESP32.

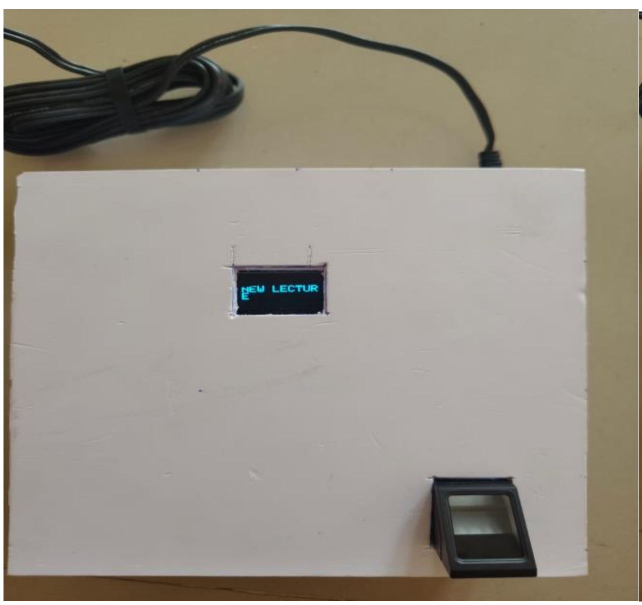
The whole project uses 5V power supply provided by a power adapter. The power is then supplied to the ESP32 board and other hardware parts, and a common ground is maintained through all circuits.

This circuitry is quite easy to make and utilizes less number of external hardware. It ensures efficient performance without any additional costs.

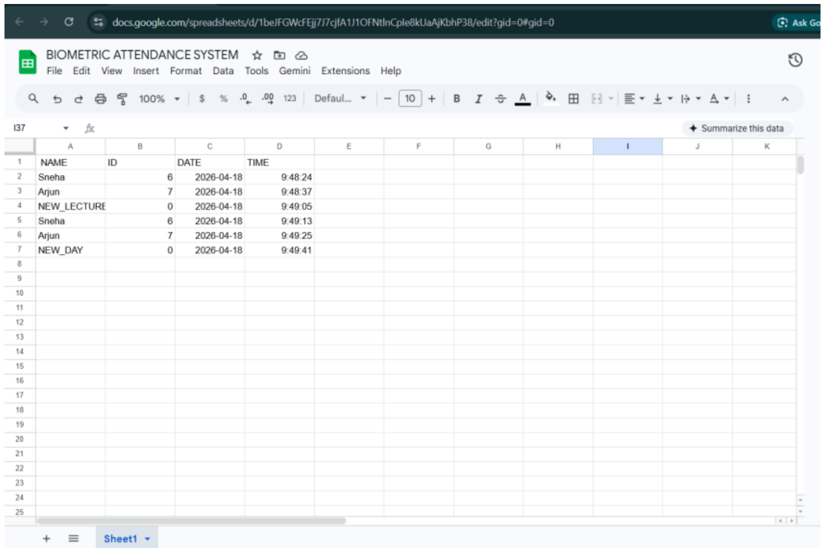
Fingerprint data are processed by the ESP32 microcontroller, which transmits the attendance details to Google Sheets through WiFi. The circuit operates on 5V regulated power supply

VI. Result and Analysis





- GOOGLE SHEET RESPONSE



- Summary

From the outputs displayed, it is clear that there was a successful deployment and operation of the biometric attendance system. Some of the system states that were depicted using the OLED display include “Connecting”, “Place Finger”, “Welcome”, “Marked”, “Already”, “New Lecture”, and “New Day”. These indicate successful interaction between the software and hardware components.

On placing one’s finger on the fingerprint reader after having been successfully registered on the system, the user is identified and their name is displayed on the screen together with attendance status. In case the attendance had already been marked, then the system alerts the user to this effect. The buttons that allow one to reset attendance to a new lecture or day are effective.

The Google sheet output reveals that the system is uploading the attendance details of each session on real-time basis. There is a successful upload of details such as name, ID, date, and time of each user’s attendance. Notably, “NEW_LECTURE” and “NEW_DAY” entries are also created on the spreadsheet.

VII. Conclusion

The suggested attendance system with biometrics integrated with cloud services will be effective and reliable for attendance purposes. Through the use of fingerprint and IoT, the proposed attendance system will ensure accuracy, efficiency, and real-time data storage.

The inclusion of ESP32 makes it possible for wireless communication. Also, with Google Sheets, it becomes possible to have an easily accessible attendance record.

Moreover, the proposed attendance system is efficient in terms of cost, scalability, and applicability.

The developed cloud-integrated attendance system with biometric thumb recognition in this project is an efficient, reliable, and accurate method for attendance purposes. With the use of fingerprints in attendance, there will be no proxy attendance cases and other forms of errors that might arise when using manual methods.

With the incorporation of ESP32 microcontroller with WiFi, it will be possible to transmit attendance records in real-time and hence access the attendance record from any place through Google Sheets. Furthermore, the OLED display facilitates user interaction and gives status information on the system.

Other additional functionalities include the use of reset buttons for reset on daily basis and per lecture basis, which makes the whole system flexible and appropriate for application in the real world scenarios. This system has been developed in a cost-effective and user-friendly manner that can accommodate more users.

In conclusion, the project has proved very successful in illustrating how the technology of biometric authentication and IoT can be effectively applied in automation of attendance systems.

VIII. REFERENCES

1. Ezeofor Chukwunazo Joseph and Georgewill Onengiye Moses (2019) developed an IoT-based student attendance monitoring system using fingerprint sensors and ESP32. The system provided accurate attendance and cloud storage but was limited by network dependency and system complexity.

2. Kotramma T. S., AzizKhan F. Pathan, Vinayaka G., Varshitha A. M., and Vikram K. Yashaswini (2023) proposed an automatic attendance system using machine learning and facial recognition techniques. The system improved automation and accuracy but required high computational power and proper lighting conditions.

3. Jawaaz Ahmad and Zahid Gulzar Khaki (2016) presented a smart classroom attendance system using sensor-based technologies such as NFC and RFID. The system improved efficiency but lacked strong authentication, allowing possible proxy attendance.

4. Sudip Misra, Anandarup Mukherjee, and Arijit Roy (2021) explained the fundamentals of IoT and its applications in real-time data communication and cloud integration. However, IoT systems depend on reliable network connectivity and security mechanisms.

5. A. K. Jain, A. Ross, and S. Prabhakar (2004) presented an overview of biometric recognition systems, highlighting fingerprint authentication as a highly secure and reliable method. However, performance may vary based on environmental conditions.

6. Roy Want (2018) discussed the concept of the Internet of Things and its applications in data communication and cloud-based systems. While effective, such systems require stable internet connectivity.

7. S. Prabhakar and A. K. Jain (2017) explored biometric authentication techniques and their applications in embedded systems. The system was reliable but lacked integration with real-time cloud communication.

8. Espressif Systems (2021) demonstrated the use of ESP32 microcontroller for IoT-based wireless communication systems. The system was efficient but dependent on network availability and configuration.

9. S. Gupta and P. Verma (2020) proposed an automated attendance system using fingerprint verification. The system reduced manual effort and improved accuracy but required proper user enrollment and maintenance.

10. A. Sharma and R. Patel (2019) developed a smart attendance system integrating biometric authentication with IoT technologies. The system improved efficiency but increased system complexity and maintenance requirements.